YIJUN BIAN

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Date of the CV: 7 November 2026

Employment

Assistant professor, Centre for Credible AI, Warsaw University of Technology (WUT), Poland — Nov 2026 to present
Researcher, Centre for Credible AI, WUT, Poland — Sept 2026 – Oct 2026
Marie Skłodowska-Curie postdoctoral fellow, University of Copenhagen (UCPH), Denmark — Aug 2024 – Jul 2026
Research fellow, National University of Singapore, Singapore — Mar 2023 – Aug 2024
Algorithm engineer, Shanghai Institute of Microsystem & Information Technology, China — Dec 2020 – Mar 2023

Education

Ph.D. in Computer Science & Technology, University of Science and Technology of China — Sept 2014 – Dec 2020
Awarded on 7 Dec 2020, locality and country: Hefei, Anhui, China
Integrated master's and doctoral programme (first two years as a master's student)
Ph.D. thesis: *Research and Applications of Diversity in Ensemble Classification*
Bachelor of Natural Sciences, Northwest A&F University (NWAFU), China — Sept 2010 – Jul 2014

Experience

Guest researcher (unpaid), UCPH, Remote — Sept 2026 – Aug 2027
Visiting scholar, Texas A&M University, United States — Nov 2018 – Apr 2019
Teaching assistant, University of Science and Technology of China (USTC), China — Mar 2016 – Jul 2016
Graduate research assistant, USTC, China — Sept 2014 – Nov 2020

Research Funding and Grants

Co-applicant & **co-lead in Denmark**, 2025 NordForsk call for Nordic research networks — NordForsk, Feb 2026
Project "Nordic BiodiversityAI Network", up to 1,999,595 NOK, selected for funding
Partners from Sweden, Finland, Norway, and Denmark, in the natural science and related engineering fields
Marie Skłodowska-Curie Actions (MSCA) Postdoctoral Fellowship awardee — European Commission, Nov 2023
Project "FairML" (CORDIS), amount € 214,934.40, competitive EU individual fellowship
FairML is selected to appear in the 'Results in Brief' section of the EC's CORDIS website — European Commission, Nov 2026
Seal of Excellence, 2022 Horizon Europe call for MSCA Postdoctoral Fellowships — European Commission, Apr 2023

Research Output (Google Scholar)

Works listed as first ***author***, equal contribution (†), and corresponding author (*)
9 published, 2 accepted; 4 peer-reviewed non-archival workshop posters, 2 under review, 4 in revision — 372 citations (cit.)

Peer-reviewed articles

[1] Huayang Liu, Mingyu Zheng, Wei Zhang, ***Yijun Bian***, Kai Wang. **SurvCancel: A longitudinal dataset and benchmark for dynamic order cancellation prediction in on-demand ride-sharing systems**, *the 40th Annual Conference on Neural Information Processing Systems (NeurIPS 2026) Evaluations and Datasets Track*, accepted.

[2] ***Yijun Bian***†* and Yujie Luo†. **Measuring model-induced discrimination via efficient fairness approximation**, *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, early access, Jul 28, 2026, doi: 10.1109/TNNLS.2026.3706648.

[3] ***Yijun Bian****. **Improving fairness with ensemble combination: Margin-dependent bounds**, *Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency (FAccT'26)*, pp. 352–381, June 25–28, 2026, Montreal, QC, Canada, doi: 10.1145/3805689.3806539.

[4] Lin Zhu†, ***Yijun Bian***†, and Lei You*. **FairSHAP: Preprocessing for fairness through attribution-based data augmentation**, *the 29th International Conference on Artificial Intelligence and Statistics (AISTATS 2026)*, May 2–5, 2026, Tangier, Morocco.

[5] Lei You*, ***Yijun Bian***, and Lele Cao. **Joint distribution-informed Shapley values for sparse counterfactual explanations**, *the 14th International Conference on Learning Representations (ICLR 2026)*, April 23–27, 2026, Rio de Janeiro, RJ, Brazil.

[6] Jinghan Huang†, Qiufeng Chen†, Pengli Zhu, ***Yijun Bian***, Nanguang Chen, Moo K. Chung, and Anqi Qiu*. **HL-HGAT: Heterogeneous graph attention network via Hodge-Laplacian operator**, *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, vol. 47, no. 12, pp. 11022–11039, Dec 2025, doi: 10.1109/TPAMI.2025.3594226.

[7] Ming Chen, *Yijun Bian*, Nanguang Chen, and Anqi Qiu*. **Orthogonal mixed-effects modeling for high-dimensional longitudinal data: An unsupervised learning approach**, *IEEE Transactions on Medical Imaging (TMI)*, vol. 44, no. 1, pp. 207–220, Jan 2025, doi: 10.1109/TMI.2024.3435855.

[8] Abhijith Sharma, *Yijun Bian*, Vatsal Nanda, Phil Munz, and Apurva Narayan*. **Vulnerability of CNNs against multi-patch attacks**, *Proceedings of the 2023 ACM Workshop on Secure and Trustworthy Cyber-Physical Systems (SaT-CPS'23)*, pp. 23–32, Apr 2023, doi: 10.1145/3579988.3585054.

[9] *Yijun Bian*, Qingquan Song, Mengnan Du, Jun Yao, Huanhuan Chen*, and Xia Hu. **Subarchitecture ensemble pruning in neural architecture search**, *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, vol. 33, no. 12, pp. 7928–7936, Dec 2022, doi: 10.1109/TNNLS.2021.3085299.

[10] *Yijun Bian* and Huanhuan Chen*. **When does diversity help generalization in classification ensembles?** *IEEE Transactions on Cybernetics (T-CYB)*, vol. 52, no. 9, pp. 9059–9075, Sept 2022, doi: 10.1109/TCYB.2021.3053165. (cit.: 135)

[11] *Yijun Bian*, Yijun Wang, Yaqiang Yao, and Huanhuan Chen*. **Ensemble pruning based on objection maximization with a general distributed framework**, *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, vol. 31, no. 9, pp. 3766–3774, Sept 2020, doi: 10.1109/TNNLS.2019.2945116. (cit.: 77)

Manuscripts in revision & Preprints

[1] *Yijun Bian**, Lei You, Yuya Sasaki, Haruka Maeda, and Akira Igarashi. **Revisiting some misconceptions and limitations in algorithmic fairness**, 2026.

[2] *Yijun Bian** and Yujie Luo. **Fast discrimination assessment for multiple non-binary sensitive attributes**, 2026.

[3] Alina Basharat, *Yijun Bian*, Ping Xu*, and Zhi Tian. **Towards trustworthy federated learning**, *arXiv preprint arXiv:2503.03684*, 2025.

[4] Abhijith Sharma†, *Yijun Bian*†*, Phil Munz, and Apurva Narayan. **Adversarial patch attacks and defences in vision-based tasks: A survey**, *arXiv preprint arXiv:2206.08304*, 2022. (cit. 63)

Selected Honours and Awards

GDC (the Global Digital Creations Holdings Limited) technology scholarship	USTC, Oct 2019
International exchange funding programme for outstanding students (Category A)	USTC, Apr 2018
Second-class academic scholarship, four times	USTC, Sept 2014, Sept 2016 – Sept 2018
First-class academic scholarship	USTC, Sept 2015
Outstanding undergraduate graduation thesis (design)	NWAFU, Jun 2014
President scholarship	NWAFU, Dec 2013
Excellence award of the undergraduate innovation forum and finding presentation	NWAFU, Jan 2013
Merit student, for three consecutive years	NWAFU, Dec 2011 – Dec 2013
First prize speciality scholarship, four times in a row	NWAFU, Mar 2011, Oct 2011 – Oct 2013

Other Key Academic Merits

- **Research supervision & leadership experience** *Co-supervision* in online collaboration (1 MSc in Canada, 2022–2023), *Co-supervision* in international collaboration (2 MEng in Singapore, 1 MSc in Denmark, 2023–2025), *Supervision* at UCPH (2 BSc, 2 MSc, 2025), *Supervision* at UCPH (3 MSc, 2025–2026), *Co-supervision* at UCPH (4 MSc, 2025–2026)
- **Teaching merits** *Teaching assistant* at USTC (2016), 2 *guest lectures* at DTU (2025–2026), 10 invited talks (2023–2026)
- **Journal reviewer** IEEE TNNLS (2016–), Neural Networks (2020–), IEEE TETCI (2021–), Journal of Supercomputing (2023–), IEEE TPAMI (2023–), Scientific Reports (2023–), IEEE TKDE (2024–), Information Sciences (2024–), IEEE TIP (2025–), IEEE TAI (2025–), IEEE JBHI (2025–), ACM TKDD (2025–)
- **Workshop reviewer** AAAI 2024 workshop on PPAI, NeurIPS 2024 workshop on M3L
- **Conference reviewer** AIES 2025, AISTATS 2026, FAccT 2026, AIES 2026

Scientific and Societal Impact

7 open-source GitHub repositories maintained for implementing the methods from my first-author research articles

- **Open-source code & official implementation for research papers** *PyEnsemble* (open-source library for ensemble learning, 2019–2020), *EPFD* (2020–2021), *SAEP* (2022), *ApproxBias* (2024–), *AssessBias* (2025–), *PyFairness* (open-source library for algorithmic fairness, 2025–), *FairML* (released resources for the MSCA project, 2025–)
- **Open-source contributions** *AutoKeras* (two merged pull requests a and b in the blocks branch, 2019), *OpenNE* (one merged pull request, 2019), *AdaNet* (Google's open-source project, one merged pull request, 2019), *Arctic Code Vault Contributor* in the 2020 GitHub Archive Programme (2020), *FairGBM* (one provided Dockerfile solution, 2023)
- **Utilising research output (own and that of others)** Falling Walls Lab Denmark (selected competitor, 2025)
- **Volunteering** GDG (Google Developer Groups) Shanghai (2021–2023)